

Part D: Recommendations Report for STADIOcare Group

Project: Predicting 30-Day Hospital Readmission Risk

Prepared for: STADIOcare Group

1. Introduction

This report presents recommendations based on the findings of Part C, which evaluated Logistic Regression (Model 1) and Random Forest (Model 2) using the Diabetes 130-US Hospitals dataset. The objective of the project is to determine whether patient and hospital encounter characteristics can identify patients at increased risk of readmission within 30 days of discharge.

The report identifies the model selected for further development, discusses potential improvements, explains how it should be adapted to STADIOcare's data, and considers whether the findings align with the literature reviewed in Part A.

2. Recommended Model: Random Forest

Based on the Part C results, Random Forest is recommended as the model to take forward for further development and validation. This recommendation is based on its performance in identifying readmissions, rather than on accuracy alone.

Both models were evaluated on the same 20,153 test observations, of which 2,151 (10.67%) represented readmissions within 30 days. Random Forest achieved a recall of 58.34%, compared with 55.14% for Logistic Regression. This represents a 3.21 percentage point increase in recall, indicating that Random Forest identified a larger proportion of actual readmissions at the default classification threshold of 0.5.

Random Forest also achieved a higher F1-score (0.2594 versus 0.2553), ROC-AUC (0.6711 versus 0.6535), and PR-AUC (0.2046 versus 0.2011). The paired bootstrap analysis estimated a ROC-AUC difference of 0.0176, with a 95% confidence interval of [0.0081, 0.0261]. This provides evidence of a positive ROC-AUC difference on the test sample.

However, Logistic Regression achieved higher accuracy (65.67% versus 64.44%), while precision was almost identical for both models (16.61% and 16.68%, respectively). The low precision of Random Forest means that many patients flagged as high risk were not readmitted within 30 days in the test data. Consequently, Random Forest should be treated as a preliminary decision-support model, not as a clinically validated prediction system. Its predictions should inform further assessment rather than replace clinical judgement.

3. Improving the Selected Model

Several improvements are recommended before Random Forest is considered for practical use.

First, the model's hyperparameters and classification threshold should be systematically optimised. The current model uses 200 trees, a maximum depth of 20, and a classification threshold of 0.5. Grid search or randomised search combined with cross-validation could assess alternative parameter settings. Threshold selection should consider the balance between identifying actual readmissions and generating unnecessary follow-up assessments. Recall, precision, PR-AUC and the number of patients flagged should be examined together.

Second, further feature engineering and data quality improvements should be investigated. Missing values, inconsistent categories and potentially informative clinical or encounter variables should be assessed. Where available, additional factors such as previous hospital admissions, discharge arrangements and follow-up care could provide further information about readmission risk. Feature importance or explainability techniques, such as SHAP, could help healthcare stakeholders understand which variables contribute to individual predictions. These explanations should not be interpreted as evidence of causation.

Third, the model should undergo additional validation. Stratified cross-validation and a separate, more recent test dataset would provide a stronger assessment of performance. Since the current dataset contains multiple encounters for some patients, future validation should consider patient-level separation to reduce the possibility of information leakage between training and test data. Performance should also be monitored over time to identify changes in patient populations, clinical practices and prediction accuracy.

4. Adapting the Model to STADIOcare's Requested Data

The current model was developed using historical US hospital encounters from 1999 to 2008. It cannot be assumed to perform reliably in STADIOcare's South African healthcare environment without adaptation and local validation.

The first step is to map the variables requested in SS1 to the features used in the existing model. STADIOcare's data should be assessed for equivalent patient demographics, diagnoses, admission and discharge information, previous encounters and other relevant clinical characteristics. Variables that are unavailable locally should be removed or replaced only where an appropriate equivalent exists.

The target outcome must also be clearly defined. STADIOcare should confirm how a readmission within 30 days will be identified, which hospitals and encounters are included, and how transfers, planned readmissions and incomplete follow-up records will be handled. The definition should be consistent across the training and evaluation data.

Next, the preprocessing pipeline must be adapted to the structure and quality of STADIOcare's data. This includes handling missing values, encoding categorical variables and ensuring that the same transformations are applied during training and prediction. The model should then be retrained using appropriately labelled local encounters and evaluated on a separate, patient-level test set. Local performance, including recall, precision, PR-AUC and calibration, should be assessed before deployment.

Finally, implementation should include appropriate access controls, privacy safeguards and clinical oversight. A pilot involving healthcare professionals should assess whether the predictions are useful and whether the volume of high-risk alerts is manageable. Deployment should proceed only after satisfactory local validation and stakeholder review.

5. Alignment with Part A Literature

The findings are broadly consistent with the literature reviewed in Part A, while also demonstrating the limitations of predicting hospital readmission.

Emi-Johnson et al. (2025), who used the Diabetes 130-US Hospitals dataset, reported differences in predictive performance across Logistic Regression, Random Forest, XGBoost and deep neural networks. Their reported ROC-AUC values were 0.642 for Logistic Regression and 0.630 for Random Forest. In the present study, Random Forest achieved a higher ROC-AUC than Logistic Regression (0.6711 compared with 0.6535). Although the direction of the comparison differs, both studies demonstrate that model performance depends on the algorithm and experimental approach. The difference also supports the need to evaluate models using consistent data and validation procedures rather than assuming that one algorithm will always perform better.

Davis et al. (2022) highlighted the value of feature engineering, cross-validation and separate testing when evaluating readmission prediction models. This supports the recommendation to optimise Random Forest and validate it using STADIOcare's own data before operational use.

Overall, the results support the use of machine learning to identify patterns associated with readmission, but they do not establish that the model can reliably predict outcomes in a different healthcare population. Local validation remains essential.

6. Conclusion

Random Forest is recommended for further development because it achieved higher recall, F1-score, ROC-AUC and PR-AUC than Logistic Regression in the Part C test results. However, its low precision and moderate discrimination mean that further improvement and validation are required. Adapting the model to STADIOcare's requested data, optimising its threshold, and conducting local clinical and technical validation are necessary steps before implementation.

Reference

- Davis, S., Zhang, J., Lee, I., Rezaei, M., Greiner, R., McAlister, F.A. and Padwal, R. (2022) 'Effective hospital readmission prediction models using machine-learned features', BMC Health Services Research, 22, article 1415. Available at: <https://doi.org/10.1186/s12913-022-08748-y> (Accessed: 25 September 2026)
- Emi-Johnson, A. et al. (2025) 'Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data'. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12085305/> (Accessed: 25 September 2026)